

Non-Bank Financial Intermediation and Systemic Risk

Research Programme Overview

Project 1 The Shadow Leverage Map

Reverse-engineering hidden NBFIs leverage from repo & derivatives data

Project 2 Mapping the Channels

Network analysis of NBFIs amplification in monetary policy transmission

Project 3 Contagion Across Borders

How NBFIs stress spills over via dollar funding & the global financial cycle

Project 4 FX Hedging as a Contagion Channel

How NBFIs currency risk management transmits global financial shocks

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1. Programme Overview

This research programme investigates the systemic risk implications of non-bank financial intermediation (NBFI) across four complementary projects. Each project addresses a distinct dimension of NBFI-related systemic risk -- hidden leverage, monetary policy amplification, cross-border contagion, and FX hedging transmission -- while sharing a common methodological toolkit based on quantile methods, network analysis, and tail-risk econometrics.

The unifying theme is that NBFI risks are fundamentally nonlinear: they are largely invisible in normal times but activate powerfully during stress. Standard mean-based econometric methods (OLS, linear VARs) miss these dynamics. All four projects therefore employ quantile-specific methods -- quantile regression, quantile connectedness (Ando, Greenwood-Nimmo & Shin 2022), Delta-CoVaR (Adrian & Brunnermeier 2016), and location-scale models (Adrian, Boyarchenko & Giannone 2019) -- to capture tail amplification and asymmetric contagion.

Project	Focus	Key Innovation
P1	Shadow leverage in repo & derivatives	Hidden Leverage Index (spectral measure)
P2	MP transmission through NBFIs	Resolves Banerjee et al. (2025) puzzle
P3	Cross-border NBFI contagion	Quantile connectedness across countries
P4	FX hedging as contagion channel	Offsetting forces / tail amplification

Project 1: The Shadow Leverage Map

Tracing Hidden Non-Bank Exposures Through Repo and Derivatives Data

NBFIs now account for nearly half of global financial assets, yet their leverage remains largely invisible to regulators. This project develops a Shadow Leverage Map that reverse-engineers leverage embedded in NBFI balance sheets using publicly available data from repo markets (OFR, ECB MMSR), derivatives reporting (DTCC), and FSB monitoring data.

Research Questions

- RQ1.** Can NBFI leverage be reverse-engineered from observable market data (repo volumes, derivatives notional, equity proxies)?
- RQ2.** How are banks and NBFIs interconnected through repo and derivatives exposures, and how do these interconnections evolve over time and during stress?
- RQ3.** Can a spectral measure of network structure (the Hidden Leverage Index, based on the largest eigenvalue of the exposure-weighted adjacency matrix) serve as a predictor of systemic events?
- RQ4.** Does hidden leverage amplify tail risk during stress? Is the bank-NBFI system more interconnected in the tails, and does high hidden leverage predict worse left-tail growth (GaR)?

Empirical Strategy

Method 1: Implied Leverage Estimation

$$\text{Implied Leverage}_i = (E_i + R_i + \text{delta} * D_i) / E_i$$

E_i = equity proxy (FSB data) R_i = repo borrowing (OFR/MMSR)

D_i = derivatives notional (DTCC), scaled by delta-equivalent

Captures economic leverage that regulatory ratios miss.

Method 2: Bipartite Bank-NBFI Network

Directed weighted network: G-SIB bank nodes and NBFI sector nodes, with edges representing repo lending and derivatives counterparty risk. Evolves quarterly.

Method 3: Hidden Leverage Index (Spectral)

$$\text{HLI}_t = \lambda_1(W_t) \text{ [largest eigenvalue of exposure matrix]}$$

Captures systemic amplification potential. Related to spectral radius in contagion models (Acemoglu, Ozdaglar & Tahbaz-Salehi 2015).

Method 4: Tail Risk Econometrics

- Quantile connectedness (tau=0.05 vs 0.50) -- is the system more interconnected in the tails?
- Tail-risk amplification: quantile regression with stress x HLI interaction.
- Growth-at-Risk (Adrian et al. 2019): does high HLI predict worse left-tail GDP growth?

Expected Contributions

1. First implied leverage measure for NBFIs from publicly observable market data.
2. Hidden Leverage Index as leading indicator -- spikes before March 2020 and Sep 2022.
3. Shadow leverage amplifies left-tail risk (GaR) but has no effect at the median.

4. Quantile connectedness confirms asymmetric contagion in bank-NBFI networks.

Data Sources & Sample

Source	Variables	Freq.	Coverage
OFR US Repo (SOFR)	Repo volumes, rates, counterparty	Daily	2014-present
ECB MMSR	Euro repo volumes by counterparty	Daily	2016-present
DTCC Swap Data	IRS/CDS notional, counterparty type	Weekly	2013-present
FSB Global Monitoring	NBFI sector AUM, leverage proxies	Annual	2002-present
BIS Locational Banking	Cross-border bank-NBFI exposures	Quarterly	2000-present
FRED / ECB SDW	GDP, financial conditions, VIX, spreads	Mixed	2000-present

Sample period: 2013Q1-2024Q4 (post-DTCC reporting). Extended to 2000 for GaR regressions using FSB annual data. Key stress episodes for validation: Taper Tantrum (May 2013), China devaluation (Aug 2015), Gilt crisis (Sep 2022), COVID-19 (Mar 2020), SVB (Mar 2023).

Literature Positioning

1. NBFI leverage measurement: Extends Jiang, Matvos, Piskorski & Seru (2024) on hidden bank losses to the NBFI sector. Complements FSB (2023) Global Monitoring with market-data approach.
2. Spectral network measures: Builds on Acemoglu et al. (2015) and Greenwood, Landier & Thesmar (2015). Adapts spectral radius to bipartite bank-NBFI setting.
3. Growth-at-Risk: Extends Adrian et al. (2019) by adding NBFI-specific predictors (hidden leverage) to financial conditions -> GDP GaR framework.

Identification & Robustness

- Endogeneity of leverage: lagged HLI values, Granger causality, IV with regulatory threshold dummies (margin call triggers).
- Measurement error: delta-equivalence factor calibrated from BIS survey; robustness across delta in [0.02, 0.10].
- Network construction sensitivity: tested across alternative edge-weighting (gross vs net, bilateral vs multilateral netting).

Project 2: Mapping the Channels

Network Analysis of Non-Bank Amplification in Monetary Policy Transmission

Building on Banerjee, Hofmann, Ng & Pinter (2025, BIS Bulletin 116), who document that OFIs amplify monetary policy transmission while ICPFs dampen it, we decompose this aggregate effect using granular network and connectedness methods. We address which OFI sub-sectors drive the amplification, through which balance-sheet channels it operates, and under what conditions it intensifies.

Hypotheses

- RQ1.** Heterogeneous amplification: procyclicality beta is highest for hedge funds, moderate for bond funds/MMFs, lowest for pension/insurance (H1).
- RQ2.** Nonlinear tail transmission: quantile IRFs at $\tau=0.05$ show 2-3x stronger responses than at $\tau=0.50$, especially for policy rate $\rightarrow$ HF returns (H2).
- RQ3.** Variance amplification: NBFI leverage increases the conditional variance of future growth outcomes, generating fatter left tails (H3).
- RQ4.** Network containment: during the 2022-23 hiking cycle, within-OFI connectedness was high but OFI-to-real-economy connectedness remained low (H4).

Empirical Strategy

Module 1: VaR-Constraint Channel (Adrian & Shin 2010, 2014)

Procyclicality of leverage by NBFI sub-sector. Fire-sale simulation: bank-only vs bank+NBFI system response to a -5% shock.

Module 2: DCC-GARCH Dynamic Correlations (Engle 2002)

Time-varying correlations between bank and NBFI sector returns. Tracks correlation spikes during stress for bank-HF, bank-MMF, bank-insurance pairs.

Module 3: Quantile VAR -- Nonlinear Transmission

$$Q_{\tau}(y_{i,t} | Y_{t-1}, \dots, Y_{t-p}) = c_i + \sum B_{ij}^{(l)}(\tau) y_{j,t-l}$$

Estimated at $\tau = \{0.05, 0.25, 0.50, 0.75, 0.95\}$.

If MP transmission through NBFIs is a tail phenomenon, IRFs at $\tau=0.05 \gg \tau=0.50$. Quantile connectedness confirms.

Module 4: Location-Scale GaR

$$\text{Location: } E[y_{t+h}] = X'\beta$$

$$\text{Scale: } \log \text{Var}[y_{t+h}] = Z'\gamma$$

If $\gamma_{\text{NBFI}} > 0$: NBFI leverage fattens the tails of future growth.

Module 5: Resolving the Banerjee et al. Puzzle

Diebold-Yilmaz decomposition shows within-OFI connectedness was high in 2022-23 but OFI-to-real-economy connectedness remained low. Banks' post-Basel III capital acted as a firewall.

Expected Contributions

1. Sub-sector decomposition: hedge funds drive amplification, pension funds dampen it.

2. Quantile connectedness reveals tail-specific MP amplification invisible to linear methods.
3. Location-scale: NBFi leverage fattens both tails (variance effect), not just shifts the mean.
4. Resolution of the 2022-23 resilience puzzle via network topology analysis.

Data Sources & Sample

Source	Variables	Freq.	Coverage
Flow of Funds (Fed Z.1)	Sector assets, liabilities, leverage	Quarterly	1980-present
ECB Inv. Fund Stats	EU fund AUM, flows, leverage by type	Quarterly	2009-present
EPFR Global	Fund flows by type, country, asset class	Monthly	2000-present
Bloomberg / Refinitiv	Sector return indices (HF, MMF, etc.)	Daily	2000-present
BIS Credit Statistics	Credit to private sector, spreads	Quarterly	1999-present
FRED	Fed funds rate, term spreads, GDP, CPI	Mixed	1960-present
Banerjee et al. (2025)	BIS Bulletin 116 replication data	Quarterly	2000-2024

Sample: 2000Q1-2024Q4 for most modules. DCC-GARCH uses daily data. Sub-sector classification: hedge funds, MMFs, bond mutual funds, equity mutual funds, ETFs, insurance, pension funds, finance companies, securitisation vehicles.

Literature Positioning

1. MP & financial intermediaries: Extends Adrian & Shin (2010, 2014) VaR-constraint channel from banks to NBFIs.
2. NBFi amplification: Builds on Banerjee et al. (2025). Sub-sector decomposition and quantile methods address their wide confidence bands.
3. Quantile VAR: Adapts Ando et al. (2022) from cross-country to cross-sector. First quantile connectedness for MP transmission.
4. Location-scale: Extends Adrian et al. (2019) GaR by modelling the scale (variance) channel separately.

Identification & Robustness

- MP shock identification: high-frequency (Gurkaynak et al. 2005); robustness with Romer & Romer (2004), Jarocinski & Karadi (2020).
- NBFi leverage endogeneity: instrumented with lagged regulatory capital ratios of connected banks.
- Quantile VAR lag selection: BIC-optimal at each quantile; robustness across p in $\{1, 2, 4\}$.
- Small-sample inference: bootstrap CIs (1000 replications) for quantile IRFs and connectedness.

Project 3: Contagion Across Borders

How NBFI Stress in One Jurisdiction Spills Over via the Global Financial Cycle

Cross-border NBFI linkages create a powerful transmission mechanism through three interconnected channels: (1) dollar funding, where NBFIs borrow in USD via repo and FX swaps; (2) portfolio rebalancing by global investment funds; and (3) the global financial cycle (Rey 2013), a common factor that NBFIs amplify through leverage and procyclical behavior.

Research Questions

- RQ1.** Is cross-border NBFI connectedness asymmetric -- dramatically higher in the left tail (stress) than at the median (normal times)?
- RQ2.** Does dollar funding stress transmit cross-border, and is this transmission nonlinear (stronger in the tails)?
- RQ3.** Are NBFI portfolio flows more volatile and more cross-border-connected than bank flows?
- RQ4.** Does NBFI penetration amplify transmission of the global financial cycle to local conditions? Robust to IV?
- RQ5.** Can a hedge fund failure cascade through prime brokerage links to generate cross-border deleveraging?

Empirical Strategy

Triple Helix: Three Contagion Channels

- Dollar funding: FX swap basis -> NBFI deleveraging -> fire sales -> cross-border (quantile regression / GaR).
- Portfolio flows: NBFI vs bank flows -> Diebold-Yilmaz connectedness on cross-country flow data.
- GFC: PCA extraction (Miranda-Agrippino & Rey 2020) -> panel regression with NBFI interaction -> IV.

Core: Cross-Country Quantile Connectedness

System: {US equity, UK equity, EU equity, JP equity, EM equity, USD funding}
 Estimated at $\tau = \{0.05, 0.50, 0.95\}$

Key prediction: Total connectedness at $\tau=0.05 \gg \tau=0.50$.

Rolling quantile connectedness tracks time-varying tail spillovers.

GFC Amplification Panel Regressions

$$y_{\{c,t\}} = \alpha_c + \beta_1 GFC_t + \beta_2 NBFI_{\{c,t\}} + \beta_3 (GFC_t \times NBFI_{\{c,t\}}) + \epsilon$$

If $\beta_3 > 0$: NBFI penetration amplifies GFC transmission.

IV: US monetary policy shocks instrument the GFC factor.

Contagion Cascade Simulation

Agent-based: US HF failure -> prime broker losses -> multi-round forced deleveraging -> cross-border loss propagation (Eisenberg-Noe + Kyle price impact).

Expected Contributions

1. Left-tail connectedness across borders is 40-60% higher than median connectedness.
2. NBFI flows are more volatile and more cross-border-connected than bank flows.
3. NBFI penetration amplifies GFC transmission (robust to IV with US MP shocks).
4. Location-scale: NBFI flows fatten the tails of EME return distributions.

Data Sources & Sample

Source	Variables	Freq.	Coverage
BIS Locational Banking	Cross-border claims by sector/country	Quarterly	2000-present
EPFR Global	Cross-border fund flows by destination	Monthly	2005-present
IMF CPIS	Portfolio positions by country pair	Annual	2001-present
BIS OTC Derivatives	FX swap/forward by currency	Semiannual	2004-present
Bloomberg	CIP basis, VIX, equity indices	Daily	2000-present
Miranda-Agrippino & Rey	Global Financial Cycle factor	Monthly	1990-present
FSB Global Monitoring	NBFI penetration ratio by country	Annual	2002-present

Sample: 2005Q1-2024Q4 for cross-country quantile connectedness. Country coverage: G20 + financial centres (US, UK, EA, JP, CH, AU, CA, KR, SG, HK, BR, MX, IN, ZA, TR, RU, CN). Separate EME sub-sample for GFC amplification tests.

Literature Positioning

1. Global financial cycle: Extends Rey (2013), Miranda-Agrippino & Rey (2020). First causal evidence of NBFI amplification of GFC via IV.
2. Cross-border contagion: Complements Forbes & Warnock (2012), Broner et al. (2013). Adds NBFI decomposition and quantile methods.
3. Dollar funding: Builds on Avdjiev et al. (2019), Eguren-Martin et al. (2024). Tests nonlinear transmission via quantile regression.
4. Network contagion: Eisenberg & Noe (2001) + Kyle (1985) price impact, extending Cont & Schaanning (2017) to cross-border setting.

Identification & Robustness

- IV for GFC factor: US MP shocks (Jarocinski & Karadi 2020), addressing reverse causality.
- NBFI penetration endogeneity: instrumented with legal origin (La Porta et al. 1998) and lagged pension reform dummies.
- Alternative connectedness: Barunik & Krehlik (2018) frequency-domain as robustness check.
- EME vs AE heterogeneity: full interaction models allowing different coefficients.

Project 4: FX Hedging as a Contagion Channel

How NBFI Currency Risk Management Transmits Global Financial Shocks

The global FX derivatives market (\$75 trillion outstanding, BIS end-2024) is driven primarily by NBFIs hedging cross-border bond investments. Combining Rey, Stavrageva & Tang (2024) on currency centrality with Nenova, Schrimpf & Shin (2025) on FX derivatives, we identify a contagion chain: equity shocks -> FX movements -> hedging cost changes -> NBFI portfolio adjustment -> bond flow reversal -> feedback.

Research Questions

- RQ1.** Do offsetting forces between equity-driven FX movements and yield curve dynamics break down in the tails? (Quantile regression of CIP basis.)
- RQ2.** Does FX hedging stress have systemic risk implications for cross-border bond flows? (Delta-CoVaR.)
- RQ3.** Does the equity-FX-CIP-bonds transmission chain activate asymmetrically during crises? (Quantile connectedness.)
- RQ4.** Do countries with higher NBFI penetration experience stronger contagion through the FX hedging channel? (Panel + IV.)

Empirical Strategy: Three Layers

Layer 1: Quantile Regression -- Offsetting Forces Test

$$Q_{\tau}(\text{CIP_basis}) = \alpha + \beta_1 \text{Equity} + \beta_2 \text{SlopeDiff} + \beta_3 \text{VIX} + \beta_4 (\text{Equity} \times \text{VIX})$$

β_4 small at $\tau=0.50$, large at $\tau=0.05$ and 0.95 .

Offsetting forces break down during stress -> amplification.

Layer 2: Delta-CoVaR (Adrian & Brunnermeier 2016)

$$\text{Delta-CoVaR} = \text{CoVaR}(\text{bond_flows} \mid \text{CIP} = \text{stressed}) - \text{CoVaR}(\text{bond_flows} \mid \text{CIP} = \text{median})$$

Large negative Delta-CoVaR = FX hedging stress amplifies bond flow tail risk.

Layer 3: Quantile Connectedness

System: {US equity, FX return, CIP basis, Bond flows}
at $\tau = \{0.05, 0.50, 0.95\}$

Transmission chain dormant at median, active during stress.

The Offsetting Forces Hypothesis

In normal times: USD appreciation raises hedging costs, but yield curve steepening makes hedged carry attractive -- partial offset. In stress: USD appreciation, curve flattening, and CIP blowout all reinforce. Testable only with quantile methods.

Expected Contributions

1. First complete model of the equity-FX-hedging-bonds contagion chain, combining Rey et al. (2024) and Nenova et al. (2025).
2. Discovery of the 'offsetting forces' mechanism: partial cancellation at the median, reinforcement in the tails.

3. First application of quantile connectedness to the FX hedging transmission chain.
4. Evidence that NBFI FX hedging maturity mismatch (3-month swaps / 10-year bonds) creates systemic rollover risk.

Data Sources & Sample

Source	Variables	Freq.	Coverage
BIS OTC Derivatives	FX swap/forward by currency & sector	Semiannual	2004-present
Bloomberg	Cross-currency basis (3M,1Y,5Y), equities	Daily	2000-present
Refinitiv	Govt bond yields (2Y,5Y,10Y), 20+ countries	Daily	2000-present
EPFR Global	Cross-border bond fund flows by country	Monthly	2005-present
IMF CPIS	Cross-border bond holdings by pair	Annual	2001-present
Rey et al. (2024)	Currency centrality, replication data	Monthly	1999-2023
Nenova et al. (2025)	FX derivatives & NBFI hedging data	Quarterly	2010-2024

Sample: 2005M1-2024M12 for quantile regressions. Currency pairs: G10 majors (USD/EUR, USD/JPY, USD/GBP, USD/AUD, USD/CAD, USD/CHF) + key EME pairs (USD/KRW, USD/MXN, USD/BRL, USD/ZAR) with liquid CIP basis data.

Literature Positioning

1. FX hedging & stability: Integrates Rey et al. (2024) on equity-FX transmission and Nenova et al. (2025) on NBFI hedging. 'Offsetting forces' is our key theoretical contribution.
2. CIP deviations: Extends Du et al. (2018), Avdjiev et al. (2019). CIP deviations as systemic risk channel, not just anomaly.
3. Delta-CoVaR: Applies Adrian & Brunnermeier (2016) to a transmission mechanism rather than an institution -- methodological novelty.
4. Maturity mismatch: Builds on Brunnermeier et al. (2009). NBFI hedging mismatch (3M swaps for 10Y bonds) analogous to bank maturity mismatch.

Identification & Robustness

- Offsetting forces: Wald test of $\beta_4(\tau=0.05) = \beta_4(\tau=0.50)$ with bootstrap p-values.
- CIP basis endogeneity: instrumented with central bank swap line announcements (exogenous supply shocks).
- Hedging demand proxy: NBFI cross-border bond holdings (CPIS) x average hedge ratio (BIS survey).
- Alternative stress measures: robust to MOVE, TED spread, financial conditions indices replacing VIX.

6. Common Methodological Toolkit

All four projects share a core econometric toolkit designed to capture nonlinear, tail-specific dynamics invisible to standard methods:

Method	What It Captures	Used In
Quantile Regression	Asymmetric effects across the distribution	P1, P2, P3, P4
Quantile VAR (QVAR)	Tail-specific shock propagation	P2, P3, P4
Quantile Connectedness	Tail spillovers between variables/countries	P1, P2, P3, P4
Delta-CoVaR	Systemic risk contribution of channels	P1, P4
Location-Scale GaR	Mean + variance effects on future growth	P2, P3
DCC-GARCH	Time-varying correlations across sectors	P2
Diebold-Yilmaz GFEVD	Directional connectedness (mean)	P2, P3
Network / spectral	Centrality, HLI, contagion matrices	P1, P3
Agent-based simulation	Cascade / fire-sale propagation	P1, P3, P4
Panel FE + IV (2SLS)	Causal identification with MP shocks	P3, P4

7. Cross-Project Linkages

The four projects are complementary. Each addresses a distinct dimension of NBFI systemic risk, but they share data infrastructure and methodology, and findings feed into each other:

- P1 (Shadow Leverage) -> P2 (MP Transmission): Implied leverage estimates serve as inputs to P2's VaR-constraint channel. Higher hidden leverage = tighter VaR constraints = amplified MP transmission.
- P2 (MP Transmission) -> P3 (Cross-Border): P2 identifies which NBFI sub-sectors amplify domestic MP; P3 traces how this propagates across borders via dollar funding and portfolio flows.
- P3 (Cross-Border) -> P4 (FX Hedging): P3's dollar funding channel (FX swap basis) is the same mechanism P4 models in detail. P4 adds the equity-FX link (Rey et al.) and yield curve offset (Nenova et al.).
- P4 (FX Hedging) -> P1 (Shadow Leverage): FX derivatives are a major source of hidden leverage (P1), and P4 shows how hedging activity creates contagion. Together: derivatives are both a leverage source AND a contagion channel.

Common finding: NBFI risks are fundamentally nonlinear. Standard mean-based methods miss tail amplification, asymmetric contagion, and crisis-activated transmission that quantile methods reveal.

8. Timeline & Deliverables

Phase	Period	Projects	Deliverables
Phase 1	Q1-Q2 2026	P1 + P2	Working papers; implied leverage dataset; QVAR estimates
Phase 2	Q3-Q4 2026	P3 + P4	Working papers; cross-country connectedness; offsetting forces
Phase 3	Q1 2027	Integration	Synthesis paper linking all projects; unified policy brief
Ongoing	Throughout	All	Conference presentations, seminar feedback, revisions

Sequencing rationale: P1 and P2 proceed in parallel (different data, complementary methods). P3 and P4 benefit from P1/P2 outputs but can begin data assembly immediately. The integration phase produces the synthesis tying all four dimensions together.

9. Policy Implications

The research programme speaks directly to ongoing regulatory debates:

1. NBFI leverage monitoring (P1): The implied leverage measure and HLI provide regulators with a prototype tool complementing FSB surveillance. Could inform NBFI leverage reporting requirements under the FSB 2023 NBFI roadmap.
2. Monetary policy & financial stability (P2): Evidence that NBFI leverage amplifies MP transmission in the tails has implications for calibrating tightening cycles. The 2022-23 resilience finding suggests post-GFC bank regulation worked but created new NBFI vulnerabilities.
3. Cross-border macroprudential coordination (P3): NBFI penetration amplifies GFC transmission, supporting arguments for cross-border macroprudential coordination (e.g., reciprocity of countercyclical capital buffers for NBFI exposures).
4. FX derivatives regulation (P4): NBFI hedging maturity mismatch (3-month swaps for 10-year bonds) creates systemic rollover risk. Supports longer-tenor hedging requirements or margin buffers calibrated to stress scenarios.
5. Data gaps (all): All projects highlight inadequate NBFI data. Supports FSB/BIS initiatives for enhanced reporting on leverage, counterparty exposures, and cross-border linkages.

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